Monetary Policy Transmission to Bond Markets: Effects of Unconventional Tools

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Abstract

The objective of this paper is to examine the effect of Bank of Canada's pandemic-era large-scale asset purchases on Canadian bond yield spreads linked to the signaling, safety/scarcity, and credit channels. Monetary policy sometimes finds itself in uncharted territory when faced with severe circumstances, such as the GFC of 07-09, or the Covid-19 Pandemic. Using a cumulative-change event-study design over seven Quantitative Easing announcement windows between March 2020 and April 2022, I estimate cumulative responses within ±5 trading days while controlling for U.S. yields, volatility, and event fixed effects. The results suggest some but weak evidence for a signaling effect once U.S. rates are controlled for, and a temporary compression in the government bond & RRB spread consistent through a portfolio-balance or scarcity channel. The credit-channel test is inconclusive because the excess bond premium series in the merged file is flat, indicating a data problem rather than an economic result.

Keywords: Quantitative Easing, monetary policy, bonds, transmission

Introduction

Globally, in 2008, the world suffered as events in the United States triggered the a financial crisis unlike anything seen since the Great Depression. Then, in 2020, a virus triggered a global pandemic that (when combined with other shocks) saw economic downturn like no other. Canada's experience during the two great financial crises of the twenty-first century demonstrates two unique economic climates. In 2008, Canada was a lesser victim of the crisis, through its commodity price collapse, but not the financial industry as with other advanced economies. It had no bank failures, no bailouts, and its recession was less severe than either the early 1980s or the early 1990s downturns. It fared demonstrably well compared to its G7 counterparts, and this despite the physical proximity to the US. The Canadian domestic banking sector, concentrated in the dominant banks (Toronto Dominion, Royal Bank of Canada, Bank of Montreal, Scotiabank, Canadian Imperial Bank of Commerce, and National Bank of Canada) and regulated by a single overarching supervisor, was never itself the source of stress. It is the responsibility of the federal government and central bank to oversee the economy, and to act during times of crises like those two major events. In Canada, there is higher faith and support in institutions to do right by society, but the expectation and pressure to not make mistakes is there. The Bank of Canada's response to the Global Financial Crisis was predictably conventional: rapid rate cuts to the effective lower bound, term repo facilities, and an outlined framework for quantitative and credit easing that was ultimately not executed. However, its response to the global pandemic was not conventional.

COVID-19 was different. Much of a nation's entire economy shutdown due to lockdowns or other restrictive policies aiming to control a virus. The pandemic shock hit the Canadian financial system through fixed-income securities trading slowing to a halt, rather than an issue of solvency. Market liquidity across all fixed-income markets almost completely dried up overnight, as a dash-for-cash mentality took hold in mid-March 2020. The bonds market is typically a cornerstone of any modern economy, and independent central banks set a standard interest rate that the economy follows as a means to meet their mandates (inflation, employment, etc.) The Bank moved its policy interest rate from 1.75% to 0.25% in a matter of weeks and launched the Government of Canada

Bond Purchase Program, the first QE program in the banks history. The latter exists because of the Zero Lower Bound: a range of interest rate values approaching zero, that restrict conventional monetary policy. When a central bank reaches this range, the operating band that they can lower or increase official rates to becomes dangerously close to the unofficial floor, 0. The the GBPP, the first of five programs, eventually accumulated roughly $340 billion in Government of Canada bonds before the Bank moved to reinvestment in October 2021 and to Quantitative Tightening in April 2022. How this translates to an effect on the bonds market (which then affects the broader economy through those channels) is immensely important to understand how Q.E. works and how to use it effectively.[1]

Literature Review

The execution was unprecedented by Canadian standards. Whether it worked, and through which channels, is less clear. The U.S. literature seems to be divided on the basic decomposition of the yield response; Gagnon et al. (2011), Bauer and Rudebusch (2014), and Krishnamurthy and Vissing-Jorgensen (2011) reach different conclusions.[2] Traditional thinking is that the pressure from expectations has a more significant downward force than scarcity (as one might expect from segmented markets theory). In the US, early LSAP studies show that bond purchases reduce long-term yields by tens of basis points but differ on whether this mainly reflects a signaling effect on the expected path of short rates or portfolio-balance effects working through term and safety premiums (Gagnon et al.; Bauer and Rudebusch; Krishnamurthy and Vissing-Jørgensen). By contrast, BoC analysis suggests QE in a small open economy may have weaker effects on long-term term premia, with GBPP announcement impacts concentrated at shorter maturities and long yields

[1] The $340 billion figure refers to total GoC bonds accumulated over the life of the program. Individual announcements committed to minimum weekly purchase rates rather than program totals, which makes the announcement-event-study design more appropriate than a program-size counterfactual.

[2] Joseph Gagnon, Matthew Raskin, Julie Remache, and Brian Sack, "The Financial Market Effects of the Federal Reserve's Large-Scale Asset Purchases"; Michael D. Bauer and Glenn D. Rudebusch, "The Signaling Channel for Federal Reserve Bond Purchases"; and Arvind Krishnamurthy and Annette Vissing-Jørgensen, "The Effects of Quantitative Easing on Interest Rates: Channels and Implications for Policy".

remaining tightly linked to global factors (Arora et al.; Diez de los Rios and Shamloo). This motivates controlling explicitly for US yields and volatility when identifying domestic signaling effects.

For Canada, two strands of work are particularly relevant. Leboeuf and Hyun construct a Canadian Excess Bond Premium series and show that it drives most of the predictive content of corporate spreads for future GDP, with a 10-basis-point increase in the EBP followed by lower GDP, lower CPI, and a flight-to-quality pattern in equity and Government of Canada yields. Arora et al. show that the Bank of Canada's Government Bond Purchase Program (GBPP), launched in 2020 to restore market functioning in the GoC bond market, lowered benchmark yields by roughly 10 to 15 basis points at announcement but that the largest effects were concentrated at shorter maturities. In line with Diez de los Rios and Shamloo, this suggests that QE in a small open economy may have weaker effects on long-term term premiums because long yields remain tightly linked to global factors, which in turn motivates controlling explicitly for US yields and volatility when identifying domestic signaling effects.

This paper takes inspiration from the Krishnamurthy and Vissing-Jorgensen framework and isolates three transmission channels to be examined with Canadian data. Each of the three channels represents a distinct avenue through which monetary policy can affect bond markets, liquidity conditions, and other macroeconomic indicators. The signaling channel rests on the unbiased expectations hypothesis, under which long rates are determined by the average of expected future short rates. The portfolio-balance or safety channel rests on the segmentation hypothesis, under which long- and short-term markets clear separately and shifts in supply within a maturity "bucket" move yields directly. The Excess Bond Premium isolates the non-default component of the corporate yield spread, making it the conceptually correct test for a credit channel that operates through risk pricing rather than expected losses. These channels will be evaluated after using Canadian data.

Data

The data set was constructed by merging daily observations from the United States Federal Reserve online database and the Bank of Canada Valet API.[3] The data for this experiment include the GoC 10-year bond yield, Canada's 3-month Treasury bill, the Real Return Bond rate, and the Excess Bond Premium. The core yield data include GoC benchmark tenors at 2-, 3-, 5-, 7-, 10-year, and long maturities, the 3-month Treasury bill, the Canadian Overnight Repo Rate Average (CORRA), and the long-maturity Real Return Bond (RRB). To isolate domestic policy shocks from global financial conditions, I also retain the 10-year U.S. Treasury yield, the 3-month U.S. Treasury bill, and the CBOE Volatility Index (VIX). Creating the term spread from the 10-year duration GoC bond minus the 3-month T-bill generates a proxy for expected future overnight rates with a term premium. The safety spread generates a proxy that shows term premiums, safety, and inflation compensation in the safe asset preferences. The third is a pre-constructed value, the Excess Bond Premium, which should capture if non-default credt premiums ease after the announcements.

The raw data set started with over 2000 observations, but dropping entries where either a proxy for non-trading days on either side of the border is missing (`goc_10yr` or `us_10yr`) leaves 745 continuous trading days. The seven event dates were selected to span the active-purchase, tapering, and tightening announcements: the initial GBPP announcement of March 27, 2020; the April 15, 2020 expansion; the October 28, 2020 recalibration; the April 21, 2021 and July 14, 2021 taper announcements; the October 27, 2021 end of QE; and the April 13, 2022 announcement of Quantitative Tightening. Stacking ±5 trading days around each event yields 11 observations per event and 77 total observations in the data.[4]

Summary statistics for the core yield-curve variables are consistent with a pandemic-era sample: the 10-year GoC yield averages 1.62% with a standard deviation of 0.93 percentage points, ranging from 0.43% at the pandemic trough to 3.66% during the 2022 normalization, and the VIX averages 24.9 with a maximum of 82.69 recorded during the March 2020 “dash-for-cash” episode.

[3] Federal Reserve Bank of St. Louis, "FRED Economic Data"; and Bank of Canada, "Valet API".

[4] Weekends and statutory holidays are implicitly excluded by the continuous-trading-day construction. Trading days on which only one of the two benchmark 10-year yields (GoC or U.S. Treasury) is missing are also dropped.

Regression Model

Following Krishnamurthy and Vissing-Jorgensen, this paper maps Canadian LSAPs into three transmission channels. The signaling channel operates through expectations of the future policy rate, as in Bauer and Rudebusch, while the portfolio-balance or safety channel reflects changes in the relative supply of safe assets across maturities and risk classes, as emphasized by Goldstein et al. and by BoC work on GBPP stock and flow effects. The credit channel is captured by the Excess Bond Premium, a non-default component of corporate spreads that has been shown for Canada to lead output and inflation and to co-move with risk-off episodes in financial markets like Leboeuf and Hyun.

In Krishnamurthy and Vissing-Jorgensen liquidity, inflation, mortgage-prepayment, and duration-risk channels complement the three core mechanisms studied here. The liquidity channel emphasizes how central-bank purchases can alleviate market dysfunction by improving trading conditions, but for Canada most of this effect appears in the form of temporary "flow" impacts on GoC yields that fade within a few days, which are already absorbed in short event windows (Arora et al. 2021). The inflation channel operates through revisions to medium and long-horizon inflation expectations, yet survey and option-based measures of Canadian inflation expectations are not available at daily frequency, and realized inflation does not move meaningfully over an eleven-day window, making this channel difficult to identify in my study. A mortgage-prepayment channel was central for US QE because the Federal Reserve purchased MBS and directly influenced refinancing incentives, but the GBPP did not target Canadian mortgage-backed securities, so this mechanism is far less relevant in the Canadian context. Finally, duration-risk effects are conceptually distinct in KVJ, but in practice they are tightly intertwined with the portfolio-balance and safety mechanisms in the long end of the GoC curve. The remaining three channels signalling, safety/scarcity, and credit, will be used in the three models after event windows and cumulative change is constructed.

Rather than regressing daily first differences, I construct cumulative changes within each event window:[5]

[5] The cumulative specification follows Arora et al. for the GBPP purchases and the broader event-study convention

$$\text{cumul } Y_{i,t} = Y_{i,t} - Y_{i,-1}$$

where $Y_{i,-1}$ is the value of the outcome on the day before event i.

Hypothesis 1: Signaling channel. LSAP announcements compress the term spread between the 10-year GoC yield and the 3-month T-bill. Made with:

$$\text{term\_spread}_t = \text{goc\_10yr}_t - \text{tbill\_3m}_t$$

Hypothesis 2: Safety/scarcity channel. LSAPs should compress the nominal-real spread between the 10-year GoC and the long-maturity RRB. Made with:[6]

$$\text{goc\_rrb\_spread}_t = \text{goc\_10yr}_t - \text{rrb\_long}_t$$

Hypothesis 3: Credit channel. A reduction in credit risk pricing around BoC announcements would compress the Excess Bond Premium.

The estimating equation for each hypothesis is:

$$\text{cumul } Y_{i,t} = \beta_0 + \sum_{k=-5,\, k\neq -1}^{5} \beta_k \mathbb{1}[\text{rel\_time}_{i,t} = k] + \sum_j \gamma_j \text{ cumul Control}_{j,i,t} + \sum_{i=2}^{7} \alpha_i D_i + \varepsilon_{i,t}$$

where D_i are event fixed effects, and the controls are the cumulative changes in the 10-year U.S. Treasury, the 3-month U.S. Treasury bill, and the VIX. I estimate the model with Newey-West standard errors using a 5-day lag truncation.

in Gagnon et al.

[6] Itay Goldstein, Jonathan Witmer, and Jing Yang, *Following the Money: Evidence for the Portfolio Balance Channel of Quantitative Easing*.

Results

A well behaved cumulative change event study should show no systematic movements in the outcome before the announcement relative to the baseline. In both the signaling and safety/scarcity specifications, the dummies pass this test. In the H1 signaling regression, the cumulative coefficients on the term spread are small in magnitude, and all four pre-event dummies rk_m5 through rk_m2 are jointly insignificant at conventional levels (F(4,57) = 0.59, $p = 0.668$).[7] In the H2 safety/scarcity regression, the corresponding pre-event dummies on the GoC RRB spread are also individually and jointly indistinguishable from zero (F(4,57) = 0.38, $p = 0.819$), which supports the interpretation that the post-announcement coefficients capture policy news rather than pre-priced trends.

The post-announcement dynamics are noticeably different in each of the three channels. For H1, the cumulative response of the term spread remains small in the five-day window, with point estimates between roughly −1 and +2 basis points and 95 percent confidence intervals that always include zero; none of the post-event dummies rk_p0 through rk_p5 is statistically significant once U.S. yields and the VIX are controlled for. This is consistent with the small-open-economy view in Bank of Canada work that Canadian long rates are tightly linked to global factors and that domestic QE may have limited incremental signaling content for the expected path of short-term rates once U.S. conditions are taken into account.[8]

By contrast, in the H2 safety/scarcity model the cumulative nominal-real spread behaves very differently. The GoC-RRB spread compresses over the subsequent three trading days, with the cumulative coefficients on rk_p2, rk_p3, and rk_p4 estimated at approximately −5.4, −6.1, and −7.2 basis points and significant at the 5 percent level, before partially reversing to about −5.3 basis points at day +5. These magnitudes, while modest, are economically meaningful in a Canadian context where the average GoC-RRB spread over the sample is about 1 percentage point with a standard deviation of roughly 0.55 percentage points.[9] The pattern–a delayed but statistically

[7] See Table 2 for regression estimates.

[8] See Arora et al. (2021) and Diez de los Rios and Shamloo (2017) for discussion of QE effects in small open economies.

[9] See the summary statistics table for levels and dispersion of the core variables.

significant compression in the nominal-real spread that fades by the end of the window–lines up with the portfolio-balance or safety interpretation emphasized by Krishnamurthy and Vissing-Jorgensen (2011) and with Bank of Canada evidence that GBPP "flow" effects on GoC yields are modest and transitory, while announcement-driven stock effects dominate the overall impact.[10]

The H3 credit-channel regression returns no useful output. The cause seems to be structural or mechanical due to the data it was fed: the Excess Bond Premium series in the event-study sample is a constant 0.60 across all 77 observations, with zero sample variance in the summary statistics table. As a result, the cumulative EBP outcome cannot be regressed on event-time dummies or controls, and the H3 column consists of zeros by construction. Given Leboeuf and Hyun's (2018) finding that the Canadian EBP is a powerful leading indicator of GDP and inflation in monthly data, this failure to identify an EBP response should be read as a data limitation of the daily merged file rather than as evidence that the QE credit channel is inoperative in Canada.

Conclusion

Using a stacked event-study design with five trading days around seven Bank of Canada announcements between March 2020 and April 2022, this paper estimates the cumulative basis-point impact of unconventional monetary policy on three spreads corresponding to the signaling, safety/scarcity, and credit channels in the Krishnamurthy-Vissing-Jorgensen (2011) framework. Specifications are estimated on 77 pooled observations with event fixed effects, controls for U.S. 10-year and 3-month yields and the VIX, and Newey-West standard errors with a five-day lag, matching the half-length of the event window. Pre-event leakage is absent in both well-identified models, which supports treating the post-announcement coefficients as the response to unexpected policy news rather than anticipated repricing.

Within this design, the term spread does not respond systematically to Canadian LSAP announcements once U.S. yields are controlled for: cumulative changes remain statistically and

[10] See Arora et al. (2021) and Goldstein, Witmer, and Yang (2018).

economically small throughout the window. This suggests that the signaling channel of QE–at least as captured by a five-day response of the 10-year-3-month term spread–is either weaker than the portfolio-balance channel in Canada or operates at lower frequencies that a short-horizon event study cannot detect, in line with Bank of Canada work emphasizing the importance of global drivers of Canadian long rates. By contrast, the GoC-RRB spread compresses by roughly 5 to 7 basis points over trading days 2 through 4, with statistically significant cumulative effects concentrated in that window and a partial reversal by day 5. This pattern is consistent with a portfolio-balance or safety mechanism in which GBPP purchases create temporary scarcity in long-duration, safe assets and compress associated premium, echoing both the KVJ portfolio-balance channel and Bank of Canada evidence that QE announcement and flow effects operate mainly through stock-driven changes in the supply of GoC bonds.

The credit-channel test remains inconclusive. The constant EBP series in the daily event-study panel prevents identification of any effect on non-default credit premium, despite separate evidence that the Canadian EBP is a key predictor of future GDP and CPI and co-moves with risk-off episodes in domestic financial markets. This reflects a failure of the data or the daily event-study specification, rather than a meaningful indication about the operation of the credit-risk channel.

Taken together, the results suggest that during the pandemic era, Bank of Canada announcements moved Canadian yield spreads primarily through a portfolio-balance or safety channel, with minor detectable incremental signaling effect at daily frequencies, and with the credit channel omitted due to design failure. Additionally, it importantly highlights the differences in the bond market, investor preferences, and liquidity of a small open economy compared to an economy like the United States. However, these findings should be interpreted within the limits of a small and heterogeneous event panel. The stacked design pools 77 observations across seven announcements that span active-purchase, tapering, end-of-QE, and quantitative-tightening regimes, so the estimated post-event coefficients represent an average response that may mask important differences between QE and QT phases. The type of result desired by the central bank may be informed by their mandates, but the type of asset purchased, and whether or not an economy is small, large, open or closed are all vital components on deciding Quantitative Easing program design. Future studies could better adapt Canadian data for all seven transmission channels, address the problem in the credit channel, and match specific purchase operation dates with appropriate market data.

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A Descriptive Statistics

Table 1: Summary Statistics of Core Variables

	count	mean	sd	min	max
term_spread	77	1.035	0.519	0.330	1.790
goc_rrb_spread	77	1.028	0.553	0.160	2.020
ebp	77	0.600	0.000	0.600	0.600
us_10yr	77	1.366	0.688	0.580	2.930
us_3m	77	0.172	0.249	-0.050	0.820
vix	77	29.633	15.244	15.010	66.040
Observations	77				

Note: All spreads and yields are in percentage levels.

B Regression Results

Table 2: Transmission Channel Estimates: Signaling, Safety, and Credit

	(1) H1: Signaling	(2) H2: Safety	(3) H3: Credit
rk_m5	0.041	-0.027	0.000
	(0.038)	(0.055)	(.)
rk_m4	0.018	-0.034	0.000
	(0.033)	(0.045)	(.)
rk_m3	0.014	-0.023	0.000
	(0.019)	(0.031)	(.)
rk_m2	0.015	-0.024	0.000
	(0.013)	(0.023)	(.)
rk_p0	-0.003	-0.006	0.000
	(0.024)	(0.016)	(.)
rk_p1	0.018	-0.029	0.000
	(0.019)	(0.021)	(.)
rk_p2	0.023	-0.054**	0.000
	(0.021)	(0.027)	(.)
rk_p3	0.006	-0.061**	0.000
	(0.023)	(0.025)	(.)
rk_p4	0.005	-0.072**	0.000
	(0.024)	(0.028)	(.)
rk_p5	-0.009	-0.053*	0.000
	(0.027)	(0.031)	(.)
Global Controls	Yes	Yes	Yes
Event Fixed Effects	Yes	Yes	Yes

Notes: Newey-West standard errors (lag 5) in parentheses.
H3 results indicate mechanical omission due to lack of variance in EBP.

C Visualizations

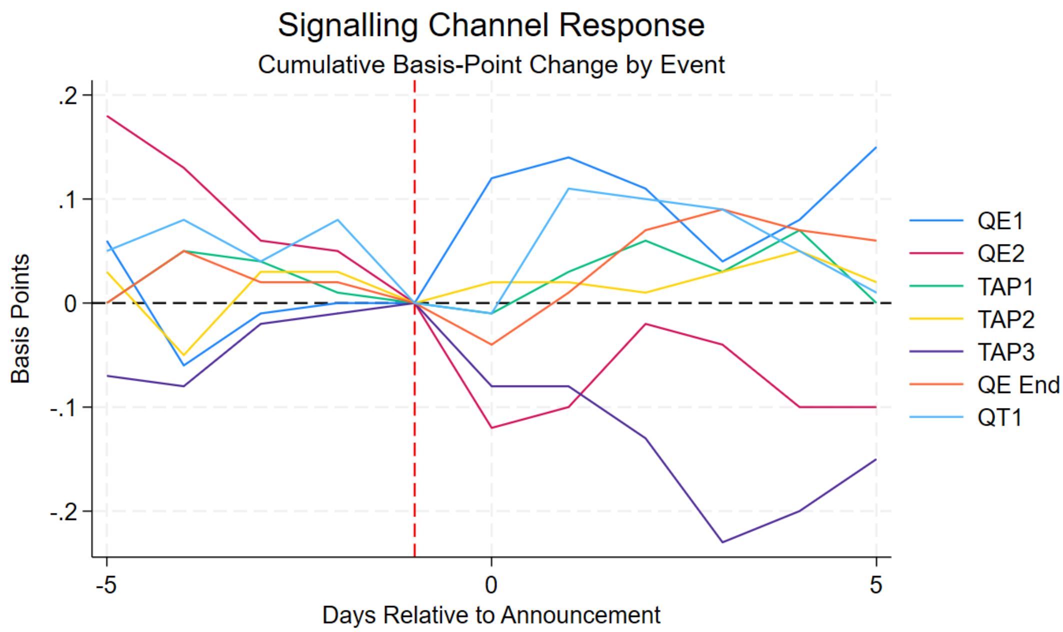

Figure 1: H1

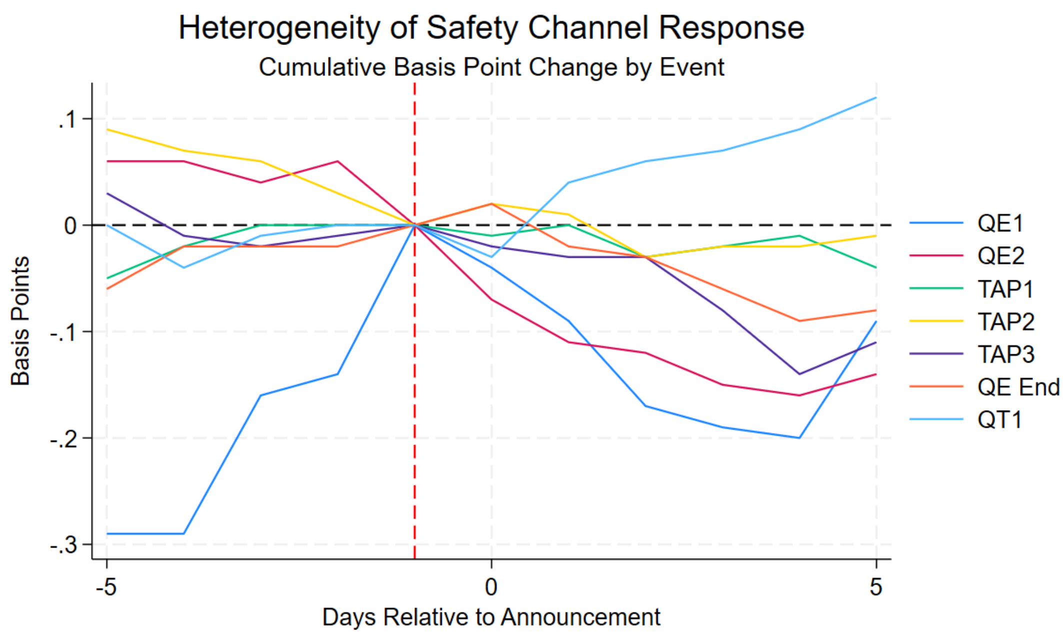

Figure 2: H2

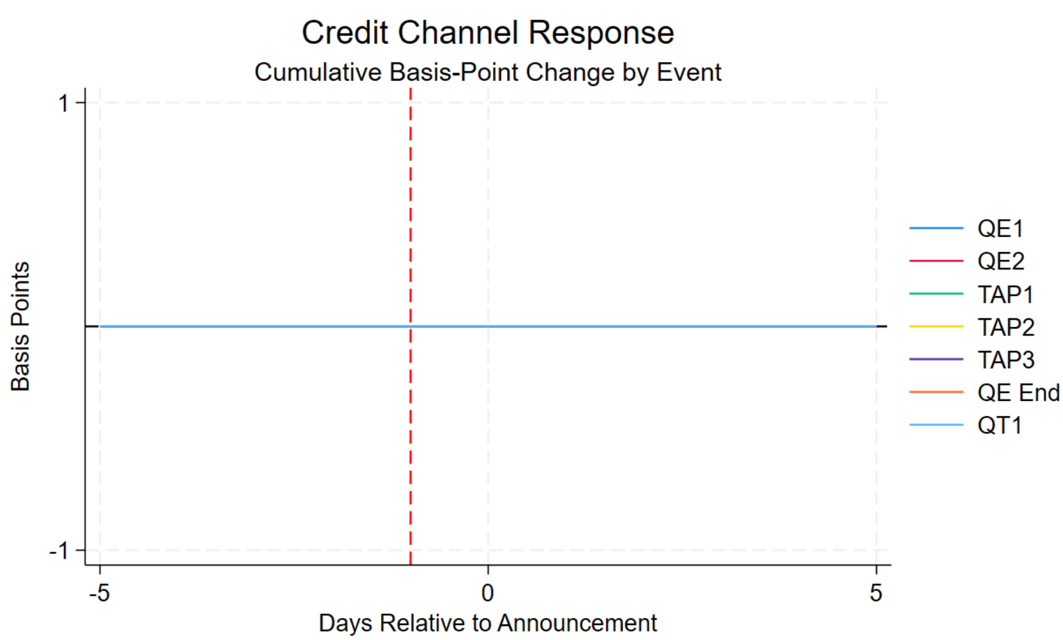

Figure 3: H3

D Log

```
------------------------------------------------------------------------------------------------

      name:  <unnamed>
       log:  D:\obsidian\Notes\Notes\Economics\Project\FinalVersion\
          BoC_KVJ_EventStudy_b00464581.log
  log type:  text
 opened on:  19 Apr 2026, 17:49:19

.
. * -----------------------------------------------------------------------
. * 1. DATA ENVIRONMENT & CLEANING
. * -----------------------------------------------------------------------
. import excel "BoC_qeQT_minimal.xlsx", sheet("Sheet1") firstrow clear
(69 vars, 775 obs)

. describe

Contains data
 Observations:           775
    Variables:            69
------------------------------------------------------------------------------------------------

Variable      Storage   Display    Value
    name         type    format    label      Variable label
------------------------------------------------------------------------------------------------

date_stata      int     %td..                 date_stata
goc_2yr         double  %14.2f                goc_2yr
goc_3yr         double  %14.2f                goc_3yr
goc_5yr         double  %14.2f                goc_5yr
goc_7yr         double  %14.2f                goc_7yr
```

goc_10yr	double	%14.2f	goc_10yr
goc_long	double	%14.2f	goc_long
rrb_long	double	%14.2f	rrb_long
goc_avg_1to3yr	double	%14.2f	goc_avg_1to3yr
goc_avg_3to5yr	double	%14.2f	goc_avg_3to5yr
goc_avg_5to10yr	double	%14.2f	goc_avg_5to10yr
goc_avg_over1~r	double	%14.2f	goc_avg_over10yr
tbill_3m	double	%14.2f	tbill_3m
corra	double	%14.2f	corra
ba_3m	double	%14.2f	ba_3m
ebp	double	%14.2f	ebp
goc_rrb_spread	double	%14.2f	goc_rrb_spread
us_10yr	double	%14.2f	us_10yr
us_3m	double	%14.2f	us_3m
sofr	double	%14.2f	sofr
fed_funds	double	%14.2f	fed_funds
corra_sofr_ba~s	double	%14.2f	corra_sofr_basis
log_settle_ba~g	double	%14.2f	log_settle_bal_lag
vix	double	%14.2f	vix
fx_usdcad	double	%14.2f	fx_usdcad
bcpi	double	%14.2f	bcpi
boc_mpc_day	byte	%14.2f	boc_mpc_day
qe_ann	byte	%14.2f	qe_ann
tap_ann	byte	%14.2f	tap_ann
qt_ann	byte	%14.2f	qt_ann
qe_ann_lag	byte	%14.2f	qe_ann_lag
tap_ann_lag	byte	%14.2f	tap_ann_lag
qt_ann_lag	byte	%14.2f	qt_ann_lag
target_rate	double	%14.2f	target_rate
v3	byte	%10.0g	v3
policy_rate	double	%14.2f	policy_rate
op_band_high	double	%14.2f	op_band_high
op_band_low	double	%14.2f	op_band_low

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bank_rate        double  %14.2f              bank_rate
prime_rate       double  %14.2f              prime_rate
settlement_ac~l  long    %14.2f              settlement_actual
settlement_bo~s  long    %14.2f              settlement_bocbs
dow              byte    %14.2f              dow
gbpp_purchase~m  byte    %14.2f              gbpp_purchased_mm
bapf_avg_yield   byte    %10.0g              bapf_avg_yield
cppp_purchase~m  byte    %14.2f              cppp_purchased_mm
pmmp_allocate~m  byte    %14.2f              pmmp_allocated_mm
gbpp_stock_mm    byte    %14.2f              gbpp_stock_mm
log_gbpp_stock   byte    %14.2f              log_gbpp_stock
cppp_stock_mm    byte    %14.2f              cppp_stock_mm
pmmp_stock_mm    byte    %14.2f              pmmp_stock_mm
log_pbpp_stock   byte    %14.2f              log_pbpp_stock
corra_spread     double  %14.2f              corra_spread
corra_eff        double  %14.2f              corra_eff
corridor_width   double  %14.2f              corridor_width
corra_lynx       double  %14.2f              corra_lynx
corra_lvts       double  %14.2f              corra_lvts
lynx_launch      byte    %14.2f              lynx_launch
days_from_lau~h  int     %14.2f              days_from_launch
bbb_aaa_spread   byte    %10.0g              bbb_aaa_spread
us_term_spread   double  %14.2f              us_term_spread
us_safety_spr~d  double  %14.2f              us_safety_spread
d_goc_rrb_spr~d  double  %14.2f              d_goc_rrb_spread
d_goc_10yr       double  %14.2f              d_goc_10yr
log_settle_bal   double  %14.2f              log_settle_bal
qe_active        byte    %14.2f              qe_active
qe_reinvest      byte    %14.2f              qe_reinvest
qt_active        byte    %14.2f              qt_active
corra_break      byte    %14.2f              corra_break
-------------------------------------------------------------------------------------------
```

```
Sorted by:
     Note: Dataset has changed since last saved.

. summarize

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
  date_stata |        775    22460.02    316.6395      21915      23009
     goc_2yr |        747    1.287644    1.294471        .15       4.28
     goc_3yr |        747    1.325087     1.23423        .18       4.23
     goc_5yr |        747    1.432945    1.063256         .3       3.84
     goc_7yr |        747    1.475649    .9883704        .32       3.69
-------------+---------------------------------------------------------
    goc_10yr |        747    1.623735    .9308941        .43       3.66
    goc_long |        747    1.954404    .7211421        .71       3.68
    rrb_long |        747    .3645248    .5256319       -.34       1.78
goc_avg_1t~r |        747    1.268568    1.292221        .14       4.27
goc_avg_3t~r |        747     1.38925    1.114773        .25       3.95
-------------+---------------------------------------------------------
goc_avg_5t~r |        747    1.550522    .9698284        .35       3.69
goc_avg_ov~r |        747    1.898461    .7808993        .52       3.71
    tbill_3m |        747    .9014458    1.247535          0       4.25
       corra |        747    .8610904    1.141405        .05       4.28
       ba_3m |        775    1.205484    1.148932         .5        4.5
-------------+---------------------------------------------------------
         ebp |        775          .6           0         .6         .6
goc_rrb_sp~d |        747     1.25921    .4789603        .16       2.11
    us_10yr |        743    1.754455    .9872759        .52       4.25
       us_3m |        743    .7995962    1.242428       -.05       4.35
        sofr |        775    .6781935    1.115519        .01       4.33
-------------+---------------------------------------------------------
   fed_funds |        775    .7069806    1.118827        .04       4.33
corra_sofr~s |        747    .1881854    .1887698       -.12          1
```

log_settle~g	292	12.30999	.4350101	5.521461	12.86612
vix	753	24.89134	8.763754	12.1	82.69
fx_usdcad	747	1.298877	.0501606	1.204	1.4496
bcpi	157	590.4925	182.8606	212.53	939.23
boc_mpc_day	775	.0322581	.1767988	0	1
qe_ann	775	.003871	.0621367	0	1
tap_ann	775	.0012903	.0359211	0	1
qt_ann	775	.0012903	.0359211	0	1
qe_ann_lag	616	.0032468	.0569339	0	1
tap_ann_lag	616	.0016234	.0402911	0	1
qt_ann_lag	616	.0016234	.0402911	0	1
target_rate	775	.9080645	1.120206	.25	4.25
v3	0				
policy_rate	775	.9080645	1.120206	.25	4.25
op_band_high	770	1.142857	1.106363	.5	4.5
op_band_low	770	.874026	1.096324	.25	4.25
bank_rate	770	1.142857	1.106363	.5	4.5
prime_rate	157	3.107643	1.11742	2.45	6.45
settlement~l	353	228170.9	38022.47	172239	291849
settlement~s	14	279358	97004.53	250	386975
dow	775	3.011613	1.412795	1	5
gbpp_purch~m	775	0	0	0	0
bapf_avg_y~d	0				
cppp_purch~m	775	0	0	0	0
pmmp_alloc~m	775	0	0	0	0
gbpp_stock~m	775	0	0	0	0
log_gbpp_s~k	775	0	0	0	0
cppp_stock~m	775	0	0	0	0

```
-------------+---------------------------------------------------------
pmmp_stock~m |        775           0           0           0          0
log_pbpp_s~k |        775           0           0           0          0
corra_spread |        747   -.0438628    .0388415         -.2      .0362
   corra_eff |        747   -.1380177    .2336504         -.8      .5724
corridor_w~h |        770    .2688312    .0660215         .25         .5
-------------+---------------------------------------------------------
  corra_lynx |        331   -.0469486     .045234         -.2        .03
  corra_lvts |        416   -.0414075    .0327306        -.12      .0362
 lynx_launch |        775    .4425806    .4970128           0          1
days_from_~h |        775   -61.98194    316.6395        -607        487
bbb_aaa_sp~d |          0
-------------+---------------------------------------------------------
us_term_sp~d |        743    .9548587    .6483639   -.7799997       2.28
us_safety_~d |        743    1.084738    .6766183   -.8800001        2.8
d_goc_rrb_~d |        585    .0004957    .0364935        -.18        .14
  d_goc_10yr |        585    .0002393    .0563649        -.26        .21
log_settle~l |        367    12.31519    .3960047    5.521461   12.86612
-------------+---------------------------------------------------------
   qe_active |        775    .5303226     .499402           0          1
 qe_reinvest |        775    .1509677    .3582486           0          1
   qt_active |        775    .2387097    .4265702           0          1
 corra_break |        775    .8477419    .3595029           0          1

.
. * Standardize date format
. capture confirm numeric variable date_stata

. if _rc == 111 {
.     gen long date_stata_num = date(date_stata, "YMD")
.     drop date_stata
.     rename date_stata_num date_stata
. }
```

```
. format date_stata %td

.
. * Structural Integrity: Drop weekends/holidays to build continuous trading
   time
. drop if missing(goc_10yr) | missing(us_10yr)
(42 observations deleted)

.
. * Time Series Setup
. sort date_stata

. gen int tday = _n

. tsset tday

Time variable: tday, 1 to 733
        Delta: 1 unit

.
. * ----------------------------------------------------------------------
. * 2. VARIABLE CONSTRUCTION (Base Levels)
. * ----------------------------------------------------------------------
. * H1: Signaling Channel (Spread between long and short bonds)
. * We construct a term spread using the 10yr GoC yield and the 3m T-Bill
. capture confirm variable term_spread

. if _rc == 111 gen term_spread = goc_10yr - tbill_3m


.
. * H2: Safety/Scarcity Channel (KVJ 2011 proxy: GoC 10yr vs Real Return Bond)
. capture confirm variable goc_rrb_spread
```

```
. if _rc == 111 gen goc_rrb_spread = goc_10yr - rrb_long

.
. * H3: Default Risk / Credit Channel (Excess Bond Premium)
. capture confirm variable ebp

. if _rc == 111 {
.     display "Warning: 'ebp' variable not found, ensure it exists in the
   dataset"
. }

.
. * Global Controls (us_10yr, us_3m, vix already exist in levels)
.
. * ---------------------------------------------------------------------
. * 3. EVENT STUDY LOGIC (+/- 5 trading days)
. * ---------------------------------------------------------------------
. local ev_dates "27mar2020 15apr2020 28oct2020 21apr2021 14jul2021 27oct2021
   13apr2022"

.
. tempfile master_data

. save "`master_data'"
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000001.tmp saved as .dta
   format

.
. local last_saved = 0

.
. * Stack windows
```

```
. forvalues i = 1/7 {
  2.     use "‘master_data’", clear
  3.     local dt : word ‘i’ of ‘ev_dates’
  4.
.     quietly summ tday if date_stata == td(‘dt’)
  5.     if r(N) == 0 continue
  6.
.     local t0 = r(mean)
  7.     keep if abs(tday - ‘t0’) <= 5
  8.
.     gen int event_id = ‘i’
  9.     gen int rel_time = tday - ‘t0’
 10.
.     tempfile ef‘i’
 11.     save "‘ef‘i’’"
 12.     local last_saved = ‘i’
 13. }
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000002.tmp saved as .dta
   format
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000003.tmp saved as .dta
   format
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000004.tmp saved as .dta
   format
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000005.tmp saved as .dta
   format
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000006.tmp saved as .dta
   format
(722 observations deleted)
```

```
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000007.tmp saved as .dta
   format
(722 observations deleted)
file C:\Users\in851325\AppData\Local\Temp\ST_8688_000008.tmp saved as .dta
   format

.
. * Append stacks together
. use "`ef`last_saved''", clear

. forvalues i = 1/7 {
  2.     if `i' == `last_saved' continue
  3.     capture confirm file "`ef`i''"
  4.     if _rc == 0 append using "`ef`i''"
  5. }

.
. * Relative-Time Loop (Omit k = -1 for causal baseline)
. foreach k of numlist -5/-2 0/5 {
  2.     local lab = cond(`k' < 0, "m" + string(abs(`k')), "p" + string(`k'))
  3.     gen byte rk_`lab' = (rel_time == `k')
  4. }

.
. global RK "rk_m5 rk_m4 rk_m3 rk_m2 rk_p0 rk_p1 rk_p2 rk_p3 rk_p4 rk_p5"

.
. * Event Fixed Effects
. tabulate event_id, generate(ev_fe)

   event_id |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |         11       14.29       14.29
```

```
          2 |         11       14.29       28.57
          3 |         11       14.29       42.86
          4 |         11       14.29       57.14
          5 |         11       14.29       71.43
          6 |         11       14.29       85.71
          7 |         11       14.29      100.00
------------+-----------------------------------
      Total |         77      100.00

. global EV_FES "ev_fe2 ev_fe3 ev_fe4 ev_fe5 ev_fe6 ev_fe7"

.
. * ----------------------------------------------------------------------
. * 3.5 CUMULATIVE VARIABLE CONSTRUCTION
. * ----------------------------------------------------------------------
. sort event_id rel_time

.
. * Updated core variables reflecting the new H1, H2, and H3
. local core_vars "term_spread goc_rrb_spread ebp us_10yr us_3m vix"

.
. foreach var of local core_vars {
  2.     bysort event_id: egen base_‘var’ = mean(cond(rel_time == -1, ‘var’,
    .))
  3.     gen cumul_‘var’ = ‘var’ - base_‘var’
  4. }

.
. * Recreate the panel_t time series to force Newey-West to run across the
   stacked data
. sort event_id rel_time
```

```
. gen int panel_t = _n

. tsset panel_t

Time variable: panel_t, 1 to 77
        Delta: 1 unit

.
. * ----------------------------------------------------------------------
. * 4. ESTIMATION & 5. POST-ESTIMATION
. * ----------------------------------------------------------------------
. di _n "===== H1: SIGNALING (Term Spread) ====="

===== H1: SIGNALING (Term Spread) =====

. newey cumul_term_spread $RK $EV_FES cumul_us_10yr cumul_us_3m cumul_vix, lag
   (5) force

Regression with Newey–West standard errors      Number of obs     =
   77
Maximum lag = 5                                F( 19,        57) =       7.04
                                               Prob > F          =     0.0000

------------------------------------------------------------------------------

              |             Newey–West
cumul_term_~d | Coefficient  std. err.      t    P>|t|     [95% conf. interval
   ]
--------------+----------------------------------------------------------------

        rk_m5 |   .0408053   .0382161     1.07   0.290    -.0357211
           .1173317
        rk_m4 |   .0183551   .0333323     0.55   0.584    -.0483916
```

						.0851019
rk_m3	.0144484	.0194484	0.74	0.461	-.0244964	.0533931
rk_m2	.0151195	.0129485	1.17	0.248	-.0108095	.0410485
rk_p0	-.0032203	.0235448	-0.14	0.892	-.050368	.0439274
rk_p1	.0181594	.0192398	0.94	0.349	-.0203676	.0566863
rk_p2	.0233979	.020687	1.13	0.263	-.0180272	.0648231
rk_p3	.0061387	.0230821	0.27	0.791	-.0400823	.0523597
rk_p4	.0051889	.0244937	0.21	0.833	-.0438589	.0542366
rk_p5	-.0087084	.0274763	-0.32	0.752	-.0637287	.0463119
ev_fe2	.0187883	.0486784	0.39	0.701	-.0786885	.1162652
ev_fe3	-.003182	.0415067	-0.08	0.939	-.0862976	.0799337
ev_fe4	-.0062679	.0409172	-0.15	0.879	-.0882032	.0756674
ev_fe5	-.0942636	.0424834	-2.22	0.030	-.179335	-.0091921
ev_fe6	.0182688	.0418256	0.44	0.664	-.0654855	.1020231
ev_fe7	.0062507	.044039	0.14	0.888	-.081936	.0944374
cumul_us_10yr	.2054001	.1482843	1.39	0.171	-.0915341	.5023343
cumul_us_3m	.4788988	.2599953	1.84	0.071	-.0417329	.9995305

```
    cumul_vix |  -.0029084   .0032335    -0.90   0.372    -.0093835
        .0035666
        _cons |   .0086294   .0375012     0.23   0.819    -.0664655
            .0837242
------------------------------------------------------------------------------


. estimates store H1

. test rk_m5 rk_m4 rk_m3 rk_m2

 ( 1)  rk_m5 = 0
 ( 2)  rk_m4 = 0
 ( 3)  rk_m3 = 0
 ( 4)  rk_m2 = 0

       F(  4,    57) =    0.59
            Prob > F =    0.6680

.
. di _n "===== H2: SAFETY/SCARCITY (RRB Spread) ====="

===== H2: SAFETY/SCARCITY (RRB Spread) =====

. newey cumul_goc_rrb_spread $RK $EV_FES cumul_us_10yr cumul_us_3m cumul_vix,
   lag(5) force

Regression with Newey West standard errors      Number of obs     =
   77
Maximum lag = 5                                 F( 19,        57) =      21.42
                                                Prob > F          =     0.0000

------------------------------------------------------------------------------
```

cumul_goc_r~d	Coefficient	Newey West std. err.	t	P>|t|	[95% conf. interval]	
rk_m5	-.0269456	.0545623	-0.49	0.623	-.1362047	.0823136
rk_m4	-.0335376	.0454425	-0.74	0.464	-.1245346	.0574593
rk_m3	-.0227067	.0311323	-0.73	0.469	-.085048	.0396346
rk_m2	-.0237215	.0233661	-1.02	0.314	-.0705114	.0230684
rk_p0	-.006154	.0157533	-0.39	0.698	-.0376994	.0253915
rk_p1	-.0293031	.0208635	-1.40	0.166	-.0710816	.0124754
rk_p2	-.0544163	.0271337	-2.01	0.050	-.1087506	-.0000821
rk_p3	-.0610649	.0247006	-2.47	0.016	-.1105271	-.0116027
rk_p4	-.0722988	.0278116	-2.60	0.012	-.1279906	-.016607
rk_p5	-.0533272	.0313764	-1.70	0.095	-.1161573	.0095028
ev_fe2	.1795335	.0617376	2.91	0.005	.0559062	.3031608
ev_fe3	.1607308	.0575113	2.79	0.007	.0455665	.2758952
ev_fe4	.2035086	.0570747	3.57	0.001	.0892184	.3177988
ev_fe5	.170791	.0477311	3.58	0.001	.075211	

```
          .2663709
       ev_fe6 |   .1597169   .0532624     3.00   0.004     .0530607
          .2663731
       ev_fe7 |   .1869174   .0561048     3.33   0.002     .0745694
          .2992653
cumul_us_10yr |   .2274577   .0989802     2.30   0.025     .0292532
   .4256621
  cumul_us_3m |   .5225016   .3026651     1.73   0.090    -.0835748
     1.128578
    cumul_vix |  -.0013074   .0033053    -0.40   0.694    -.0079262
       .0053114
        _cons |  -.1515997   .0674018    -2.25   0.028    -.2865696
           -.0166299
------------------------------------------------------------------------------


. estimates store H2

. test rk_m5 rk_m4 rk_m3 rk_m2

 ( 1)  rk_m5 = 0
 ( 2)  rk_m4 = 0
 ( 3)  rk_m3 = 0
 ( 4)  rk_m2 = 0

       F(  4,    57) =    0.38
            Prob > F =    0.8188

.
. di _n "===== H3: CREDIT (Excess Bond Premium) ====="

===== H3: CREDIT (Excess Bond Premium) =====
```

```
. newey cumul_ebp $RK $EV_FES cumul_us_10yr cumul_us_3m cumul_vix, lag(5)
   force

Regression with Newey West standard errors      Number of obs     =
   77
Maximum lag = 5                                  F(  0,        57) =          .
                                                 Prob > F          =          .

-------------------------------------------------------------------------------

             |             Newey West
   cumul_ebp | Coefficient  std. err.      t    P>|t|     [95% conf. interval
      ]
-------------+-----------------------------------------------------------------

       rk_m5 |          0  (omitted)
       rk_m4 |          0  (omitted)
       rk_m3 |          0  (omitted)
       rk_m2 |          0  (omitted)
       rk_p0 |          0  (omitted)
       rk_p1 |          0  (omitted)
       rk_p2 |          0  (omitted)
       rk_p3 |          0  (omitted)
       rk_p4 |          0  (omitted)
       rk_p5 |          0  (omitted)
      ev_fe2 |          0  (omitted)
      ev_fe3 |          0  (omitted)
      ev_fe4 |          0  (omitted)
      ev_fe5 |          0  (omitted)
      ev_fe6 |          0  (omitted)
      ev_fe7 |          0  (omitted)
cumul_us_10yr |          0  (omitted)
  cumul_us_3m |          0  (omitted)
```

```
    cumul_vix |          0  (omitted)
        _cons |  -2.38e-08          .         .       .             .
           .
--------------------------------------------------------------------------------


. estimates store H3

. test rk_m5 rk_m4 rk_m3 rk_m2

 ( 1)  rk_m5 = 0
 ( 2)  rk_m4 = 0
 ( 3)  rk_m3 = 0
 ( 4)  rk_m2 = 0
       Constraint 1 dropped
       Constraint 2 dropped
       Constraint 3 dropped
       Constraint 4 dropped

       F(  0,    57) =       .
            Prob > F =         .

.
. * ----------------------------------------------------------------------
. * 6. NATIVE STATA COEFPLOTS (Flattened to defeat parser bugs)
. * ----------------------------------------------------------------------
. capture drop k_axis b_val ci_hi ci_lo

. gen k_axis = .
(77 missing values generated)

. gen b_val = .
(77 missing values generated)
```

```
. gen ci_hi = .
(77 missing values generated)

. gen ci_lo = .
(77 missing values generated)

.
. local k_vals "-5 -4 -3 -2 0 1 2 3 4 5"

.
. foreach m in H1 H2 H3 {
  2.     est restore ‘m’
  3.
.     local idx = 1
  4.     foreach v of global RK {
  5.         local k : word ‘idx’ of ‘k_vals’
  6.         quietly replace k_axis = ‘k’ in ‘idx’
  7.         quietly replace b_val = _b[‘v’] in ‘idx’
  8.         quietly replace ci_hi = _b[‘v’] + 1.96 * _se[‘v’] in ‘idx’
  9.         quietly replace ci_lo = _b[‘v’] - 1.96 * _se[‘v’] in ‘idx’
 10.         local idx = ‘idx’ + 1
 11.     }
 12.
.     * ONE SINGLE LINE: No slashes, no continuations, no nodraw.
.     twoway (rcap ci_hi ci_lo k_axis in 1/10, lcolor(gs7)) (scatter b_val
   k_axis in 1/10, mcolor(navy) msize(medium)), yline(0,
> lcolor(black) lpattern(solid)) xline(-1, lcolor(red) lpattern(dash)) xlabel
   (-5(1)5) xtitle("Trading days relative to announceme
> nt (k = -1 omitted)") ytitle("Coefficient (pp)") title("‘m’: Dynamic Effects
    (7 events pooled)") legend(off)
 13.
.     graph export "‘m’_coefplot.png", replace
```

```
 14.     graph close
 15. }
(results H1 are active now)
file H1_coefplot.png saved as PNG format
(results H2 are active now)
file H2_coefplot.png saved as PNG format
(results H3 are active now)
file H3_coefplot.png saved as PNG format

.
. * desc  stats
. local summ_vars "term_spread goc_rrb_spread ebp us_10yr us_3m vix"

.
. estpost summarize ‘summ_vars’, listwise

             |  e(count)   e(sum_w)    e(mean)     e(Var)      e(sd)     e(min
                )      e(max)     e(sum)
-------------+---------------------------------------------------------------------------

 term_spread |        77         77   1.035065   .2691806   .5188262
     .33        1.79        79.7
goc_rrb_sp~d |        77         77   1.028182    .305323   .5525604
    .16        2.02       79.17
         ebp |        77         77         .6          0          0
            .6          .6       46.2
     us_10yr |        77         77   1.366364   .4731077   .6878282
         .58        2.93     105.21
       us_3m |        77         77   .1718182   .0618388    .248674
          -.05         .82      13.23
         vix |        77         77   29.63338   232.3943   15.24448
            15.01       66.04     2281.77
```

```
. esttab using "Descriptive_Stats.tex", replace ///
>     cells("count(fmt(0)) mean(fmt(3)) sd(fmt(3)) min(fmt(3)) max(fmt(3))")
   ///
>     label booktabs nonumber title("Summary Statistics of Core Variables")
   ///
>     addnotes("Note: All spreads and yields are in percentage levels.")
(output written to Descriptive_Stats.tex)

.
. * the REG RESULTS
. esttab H1 H2 H3 using "Regression_Results.tex", replace ///
>     b(3) se(3) star(* 0.10 ** 0.05 *** 0.01) ///
>     label booktabs alignment(c) ///
>     title("Transmission Channel Estimates: Signaling, Safety, and Credit")
   ///
>     mtitles("H1: Signaling" "H2: Safety" "H3: Credit") ///
>     drop(_cons) ///
>     indicate("Global Controls = cumul_us_10yr cumul_us_3m cumul_vix" "Event
   Fixed Effects = ev_fe*", labels("Yes" "No")) ///
>     addnotes("Notes: Newey-West standard errors (lag 5) in parentheses." "H3
    results indicate mechanical omission due to lack o
> f variance in EBP.") ///
>     noobs nonotes compress
(output written to Regression_Results.tex)

.
. xtline cumul_term_spread, overlay ///
>     t(rel_time) i(event_id) ///
>     title("Signalling Channel Response") ///
>     subtitle("Cumulative Basis-Point Change by Event") ///
>     xtitle("Days Relative to Announcement") ///
>     ytitle("Basis Points") ///
>     xline(-1, lcolor(red) lpattern(dash)) ///
```

```
>     yline(0, lcolor(black)) ///
>     legend(label(1 "QE1") label(2 "QE2") label(3 "TAP1") ///
>            label(4 "TAP2") label(5 "TAP3") label(6 "QE End") label(7 "QT1"))

.
. * This line generates the actual file
. graph export "H1_SpaghettiPlot.png", replace
file H1_SpaghettiPlot.png saved as PNG format

.
.
. * testing viz
. xtline cumul_goc_rrb_spread, overlay ///
>     t(rel_time) i(event_id) ///
>     title("Safety Channel Response") ///
>     subtitle("Cumulative Basis-Point Change by Event") ///
>     xtitle("Days Relative to Announcement") ///
>     ytitle("Basis Points") ///
>     xline(-1, lcolor(red) lpattern(dash)) ///
>     yline(0, lcolor(black)) ///
>     legend(label(1 "QE1") label(2 "QE2") label(3 "TAP1") ///
>            label(4 "TAP2") label(5 "TAP3") label(6 "QE End") label(7 "QT1"))

.
. * This line generates the actual file
. graph export "H2_SpaghettiPlot.png", replace
file H2_SpaghettiPlot.png saved as PNG format

.
.
. xtline cumul_ebp, overlay ///
>     t(rel_time) i(event_id) ///
>     title("Credit Channel Response") ///
```

```
>     subtitle("Cumulative Basis-Point Change by Event") ///
>     xtitle("Days Relative to Announcement") ///
>     ytitle("Basis Points") ///
>     xline(-1, lcolor(red) lpattern(dash)) ///
>     yline(0, lcolor(black)) ///
>     legend(label(1 "QE1") label(2 "QE2") label(3 "TAP1") ///
>            label(4 "TAP2") label(5 "TAP3") label(6 "QE End") label(7 "QT1"))

.
. * This line generates the actual file
. graph export "H3_SpaghettiPlot.png", replace
file H3_SpaghettiPlot.png saved as PNG format

. * ---------------------------------------------------------------------------
. * CLEANUP
. * ---------------------------------------------------------------------------
. graph close _all

. log close
      name:  <unnamed>
       log:  D:\obsidian\Notes\Notes\Economics\Project\FinalVersion\
          BoC_KVJ_EventStudy_b00464581.log
  log type:  text
 closed on:  19 Apr 2026, 17:49:22
-------------------------------------------------------------------------------------------
```